RMI: A Machine-Native Framework for Autonomous AI System Generation

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Abstract

We present RMI (Recursive Machine Intelligence), a novel framework designed fundamentally for AI systems to generate, compose, and optimize other AI systems—without human intervention. Unlike conventional machine learning frameworks that encode human cognitive biases through natural language abstractions and interfaces optimized for human developers, RMI (Recursive Machine Intelligence) inverts this paradigm by providing machine-native primitives, binary-first communication protocols, and compositional algebraic structures that AI agents can reason over directly. The framework introduces several key innovations: (1) a typed intermediate representation (IR) for machine-manipulable program synthesis, (2) formal mutation and crossover operators enabling evolutionary code generation, (3) a hybrid neurosymbolic reasoning engine that adaptively combines continuous and discrete inference, (4) a maximally efficient binary protocol for inter-agent communication with direct tensor sharing, and (5) self-describing ontological components for autonomous capability discovery. We demonstrate that RMI (Recursive Machine Intelligence) enables autonomous agents to design neural architectures, synthesize programs, and collaborate on complex tasks with significantly reduced human involvement. Our implementation in Rust achieves both memory safety and performance, with comprehensive support for neural, symbolic, and neurosymbolic AI paradigms.

Keywords: Autonomous AI Systems, Program Synthesis, Neurosymbolic AI, Multi-Agent Systems, Machine Learning Frameworks, Intermediate Representations, Evolutionary Computation

1 Introduction

The proliferation of machine learning frameworks over the past decade—including PyTorch [1], TensorFlow [2], JAX [3], and others—has dramatically accelerated AI research and deployment. However, these frameworks share a fundamental assumption: they are designed for *human* developers. This design philosophy manifests in interfaces optimized for human cognition: Python-based APIs, natural language documentation, sequential programming models, and debugging tools that present information visually.

As AI systems become increasingly capable of autonomous operation, a fundamental question arises: *What would a machine learning framework look like if it were designed for AI systems rather than humans?*

This question motivates RMI (Recursive Machine Intelligence), a framework that inverts the traditional human-centric design paradigm. Rather than providing abstractions that humans find intuitive, RMI (Recursive Machine Intelligence) provides primitives that machines can manipulate directly and efficiently. The key insight is that AI agents do not benefit from—and are often hindered by—abstractions designed for human cognition. String parsing, dynamic typing, reflection, and natural language interfaces all impose computational overhead and ambiguity that impede machine reasoning.

RMI (Recursive Machine Intelligence) addresses this through several innovations:

1. **Machine-Native Intermediate Representation**: A strongly-typed IR that AI systems can generate, analyze, and transform through formal operations rather than text manipulation.
2. **Algebraic Compositionality**: All components compose through formal algebraic structures, enabling machines to reason about program equivalence, optimization, and synthesis.
3. **Binary-First Protocols**: Maximally efficient inter-agent communication using MessagePack serialization with LZ4 compression, eliminating the overhead of human-readable formats.
4. **Neurosymbolic Integration**: Seamless bridging of neural (continuous) and symbolic (discrete) reasoning, allowing agents to leverage both paradigms as appropriate.
5. **Self-Describing Ontologies**: Rich metadata enabling autonomous discovery of capabilities, constraints, and composition rules.

The remainder of this paper is organized as follows. Section 2 discusses related work and positions RMI (Recursive Machine Intelligence) relative to existing frameworks. Section 3 presents the system architecture and core design principles. Section 4 details the key technical contributions. Section 5 describes the implementation. Section 6 provides evaluation results. Section 7 concludes with future directions.

2 Related Work

2.1 Machine Learning Frameworks

Modern ML frameworks have evolved through several generations. First-generation frameworks like Theano [4] provided symbolic computation graphs requiring explicit compilation. Second-generation frameworks including TensorFlow [2] introduced production-ready distributed computing but retained static graph semantics. Third-generation frameworks like PyTorch [1] pioneered dynamic computation graphs through eager execution, dramatically improving developer experience.

Recent frameworks have explored various specialized directions: JAX [3] emphasizes functional transformations and composable program transformations; Flax and Haiku provide functional neural network libraries; MLX targets Apple Silicon. However, all these frameworks maintain the fundamental assumption of human developers as the primary users.

2.2 Program Synthesis

Program synthesis has a rich history spanning deductive synthesis [5], inductive programming [6], and neural program synthesis [7, 8]. Recent advances in large language models have demonstrated impressive code generation capabilities [9, 10], but these approaches generate text that must be parsed and compiled, introducing inefficiency and potential errors.

Genetic programming [11] and evolutionary algorithms [12] operate on structured representations, closer to our approach. However, traditional genetic programming lacks the formal type systems and algebraic properties that enable principled composition.

2.3 Multi-Agent Systems

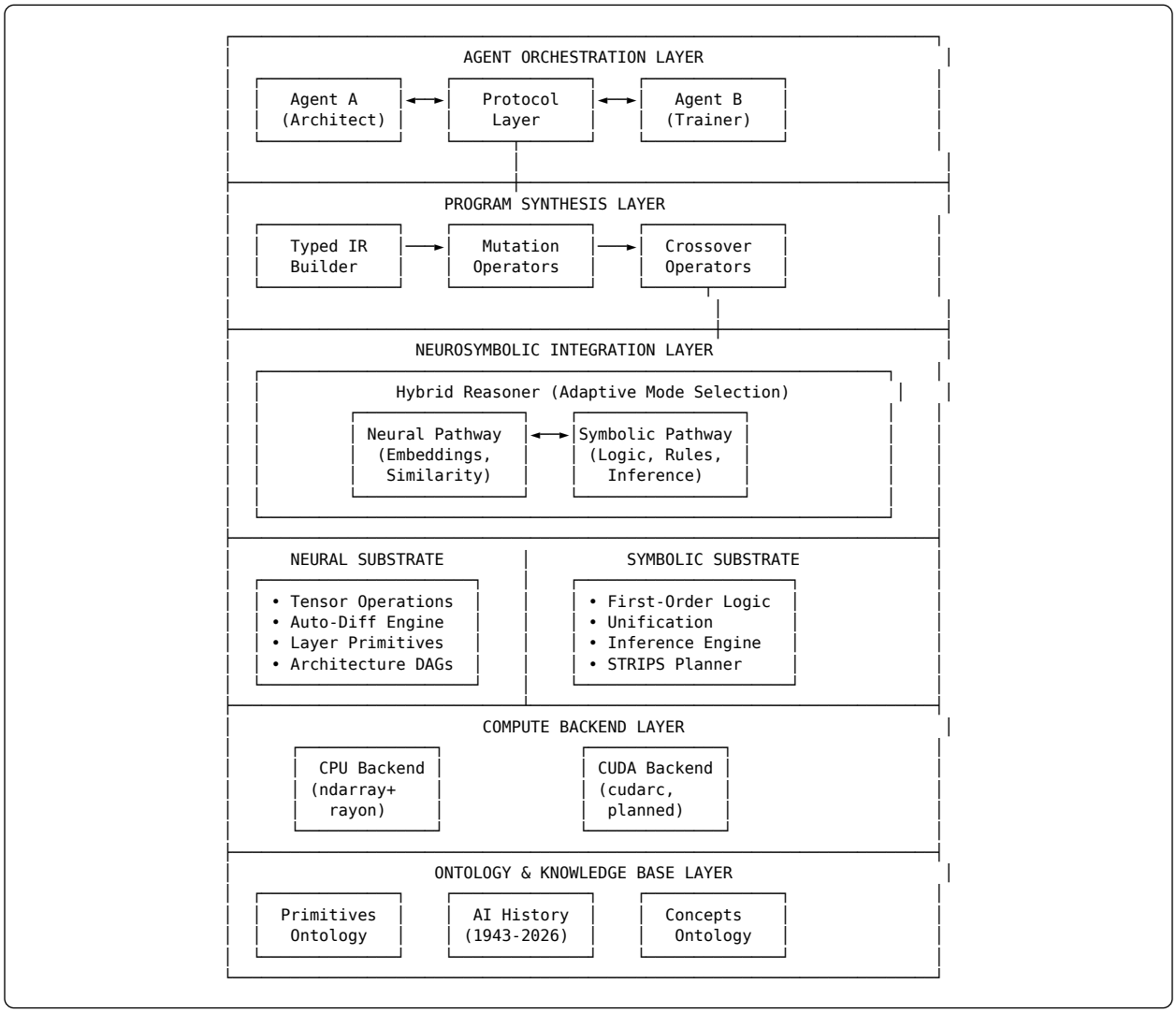
Multi-agent AI systems have been explored extensively, from early work on distributed AI [13] to modern frameworks for agent coordination [14]. Recent work on LLM-based agents [15, 16] demonstrates emergent collaborative capabilities, but communication typically occurs through natural language, limiting efficiency.

2.4 Neurosymbolic AI

The integration of neural and symbolic approaches has gained significant attention [17, 18, 19]. Systems like Neural Theorem Provers [20], DeepProbLog [21], and differentiable reasoning approaches [22] bridge these paradigms. RMI (Recursive Machine Intelligence) builds on this foundation by providing a unified framework where neurosymbolic integration is a first-class primitive.

3 System Architecture

RMI (Recursive Machine Intelligence) is organized as a layered architecture where each layer provides abstractions that higher layers can compose and reason over. Listing 1 illustrates the overall system structure.



Listing 1: RMI (Recursive Machine Intelligence) layered architecture. Arrows indicate primary data flow and communication paths between components.

3.1 Design Principles

The architecture embodies several key principles:

Zero Human Abstraction Tax. Every abstraction in the system exists to serve machine reasoning, not human comprehension. This eliminates string parsing, dynamic dispatch overhead, and runtime type checking that characterize human-oriented frameworks.

Compositional Algebraic Structure. Components compose through well-defined algebraic operations. This enables formal reasoning about program equivalence, optimization opportunities, and synthesis constraints.

Self-Description. Every component carries metadata sufficient for autonomous discovery and reasoning. Agents can query capabilities, constraints, and composition rules without external documentation.

Binary Efficiency. Inter-agent communication uses maximally efficient binary protocols. Where human frameworks might use JSON over HTTP, RMI (Recursive Machine Intelligence) uses MessagePack with LZ4 compression over direct memory channels.

3.2 Component Overview

Agent Orchestration Layer: Manages autonomous AI agents, their lifecycle, goals, and coordination. Agents communicate through the protocol layer and can discover each other's capabilities.

Program Synthesis Layer: Provides the typed IR and genetic operators (mutation, crossover) for evolutionary program generation. This layer enables agents to synthesize and evolve code.

Neurosymbolic Integration Layer: Bridges neural and symbolic computation, providing hybrid reasoning capabilities that adaptively select between continuous and discrete inference.

Neural Substrate: Provides differentiable computation primitives, automatic differentiation, and neural architecture building blocks.

Symbolic Substrate: Provides first-order logic, unification, inference engines, and planning capabilities.

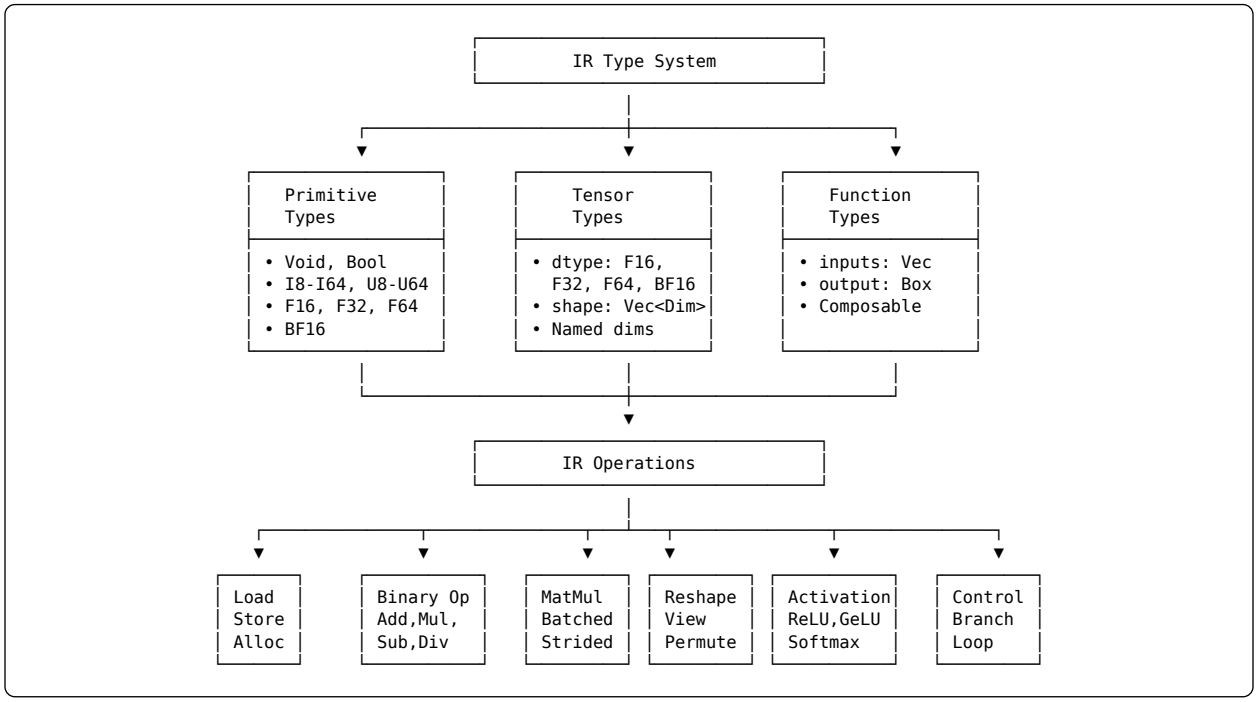
Compute Backend Layer: Abstracts hardware execution, currently supporting CPU (with planned CUDA support).

Ontology & Knowledge Base: Stores machine-readable knowledge including primitive specifications, AI history, and conceptual ontologies.

4 Technical Contributions

4.1 Machine-Native Typed Intermediate Representation

A central contribution of RMI (Recursive Machine Intelligence) is the typed intermediate representation (IR) designed for machine manipulation rather than human readability. Unlike source code that must be parsed from text, the IR exists as structured data that AI systems can directly traverse, analyze, and transform.



Listing 2: IR type system and operation hierarchy. The type system ensures static verification while operations cover neural network primitives.

The IR type system (Listing 2) provides:

Static Type Safety. All operations are statically typed, enabling compile-time verification of tensor shapes, data types, and function signatures. This eliminates runtime shape errors that plague dynamic frameworks.

Dimension Polymorphism. Tensor types support both concrete dimensions and symbolic dimensions, enabling generic functions over tensor shapes:

```
IRType::Tensor {
    dtype: PrimitiveType::F32,
    shape: vec![Dimension::Named("batch"),
                Dimension::Concrete(784)],
}
```

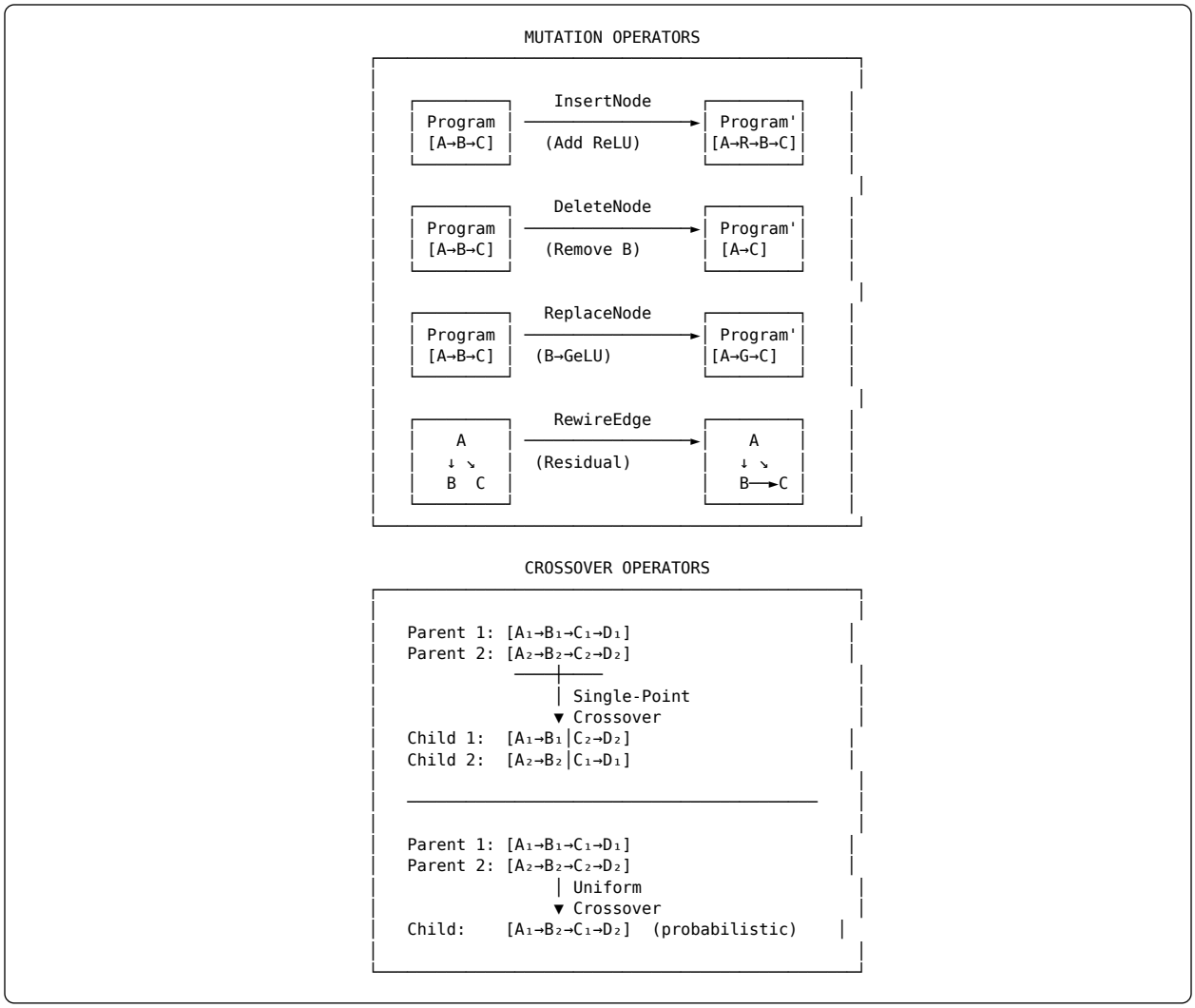
Operation Set. The IR supports operations sufficient for neural network construction: memory operations (load, store, allocate), binary operations (add, multiply, etc.), matrix operations (matmul with batching and transposition), tensor reshaping, activations, and control flow.

Structural Hashing. Programs can be hashed based on structure rather than identity, enabling deduplication of semantically equivalent programs:

```
fn structural_hash(&self) -> u64 {
    let mut hasher = DefaultHasher::new();
    // Hash based on semantic structure, not memory layout
    self.ops.iter().for_each(|op| op.hash(&mut hasher));
    hasher.finish()
}
```

4.2 Evolutionary Program Synthesis

RMI (Recursive Machine Intelligence) provides formal mutation and crossover operators for evolutionary program generation. Unlike traditional genetic programming that operates on untyped tree structures, our operators maintain type safety and semantic validity.



Listing 3: Mutation and crossover operators for evolutionary program synthesis. All operators maintain type safety.

4.2.1 Mutation Operators

The mutation system (Listing 3 top) provides four primitive operators:

InsertNode: Inserts a new operation at a random valid position, respecting type constraints:

```
Mutation::InsertNode {
    function: func_id,
    position: 3,
    node: IRNode::Activation(ActivationKind::GeLU)
}
```

DeleteNode: Removes an operation while maintaining graph connectivity. If removing a node would break data dependencies, the operator rewires edges to maintain validity.

ReplaceNode: Substitutes one operation with another of compatible type signature, enabling exploration of equivalent computations.

RewireEdge: Changes data dependencies without altering operations, enabling architectural variations like skip connections.

4.2.2 Crossover Operators

Crossover (Listing 3 bottom) combines genetic material from two parent programs:

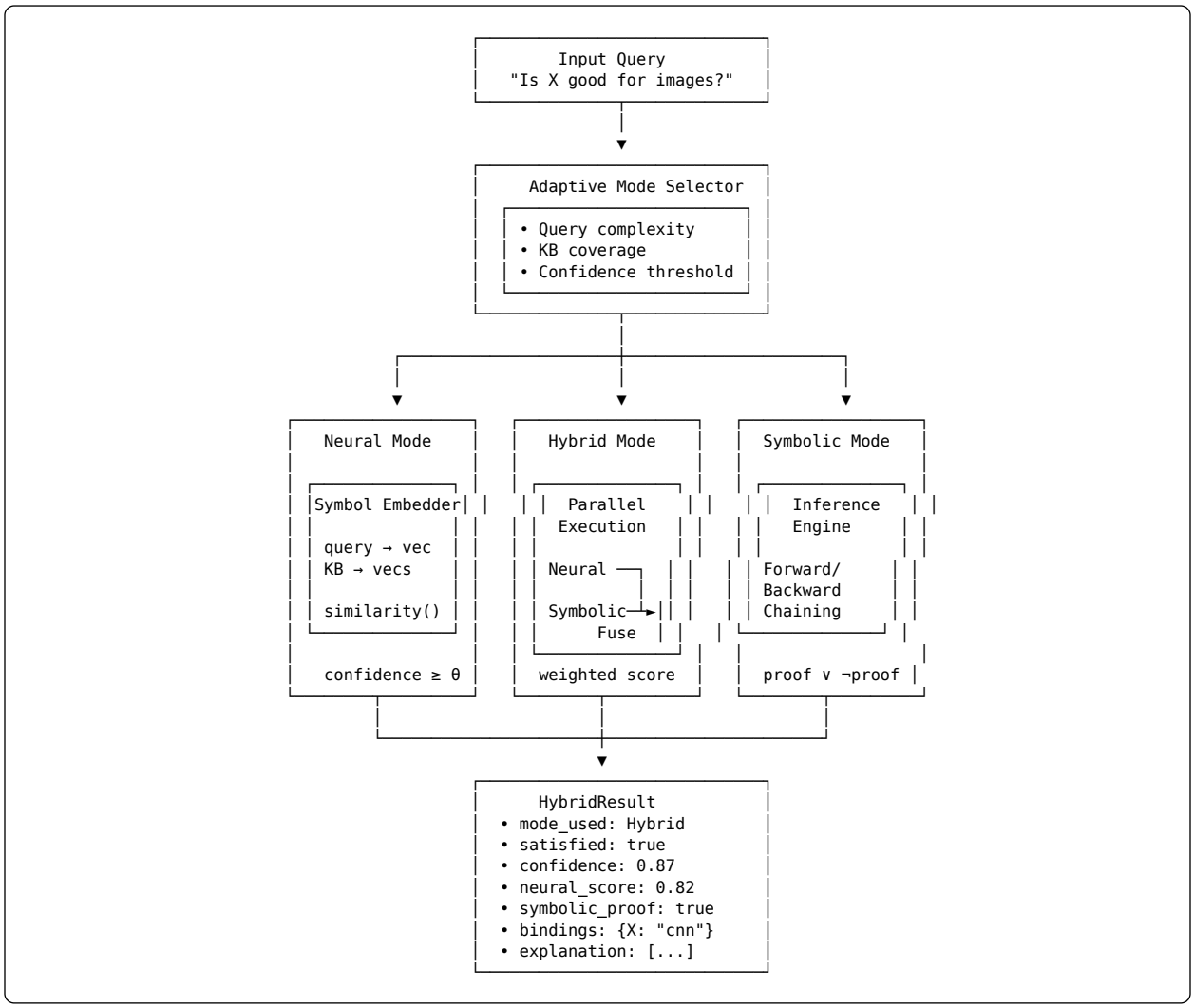
Single-Point Crossover: Selects a compatible cut point in both parents and swaps suffixes, producing two children.

Uniform Crossover: For each position, probabilistically selects from either parent, enabling fine-grained recombination.

Both operators maintain type safety by validating that cross-points have compatible types and that resulting programs pass type checking.

4.3 Hybrid Neurosymbolic Reasoning

A key innovation of RMI (Recursive Machine Intelligence) is the seamless integration of neural and symbolic reasoning through the HybridReasoner component. This enables AI agents to leverage continuous representations (neural networks, embeddings) and discrete representations (logic, rules) within a unified framework.



Listing 4: Hybrid neurosymbolic reasoning pipeline. The adaptive mode selector routes queries to neural, symbolic, or hybrid execution paths based on problem characteristics.

4.3.1 Reasoning Modes

The HybridReasoner (Listing 4) supports four modes:

Neural Mode: Converts symbolic queries to embeddings and performs similarity-based reasoning. Effective for fuzzy matching and when the knowledge base lacks explicit rules.

Symbolic Mode: Uses traditional logical inference (forward/backward chaining). Provides provable answers with explicit derivation chains.

Hybrid Mode: Executes both pathways in parallel and fuses results using configurable weights:

$$\text{score}_{\text{hybrid}} = \alpha \cdot \text{score}_{\text{neural}} + (1 - \alpha) \cdot \text{score}_{\text{symbolic}}$$

where α is the neural weight parameter.

Adaptive Mode: Automatically selects the most appropriate mode based on query characteristics, knowledge base coverage, and runtime statistics.

4.3.2 Symbol Embedding

The SymbolEmbedder converts symbolic structures (terms, predicates, formulas) into continuous vector representations:

```
impl SymbolEmbedder {
    fn embed_predicate(&mut self, pred: &Predicate) -> Vec<f32> {
        let name_emb = self.embed_symbol(&pred.name);
        let args_emb = pred.args.iter()
            .map(|t| self.embed_term(t))
            .reduce(|a, b| self.aggregate(&a, &b))
            .unwrap_or_else(|| vec![0.0; self.config.embedding_dim]);

        self.combine_embeddings(&name_emb, &args_emb)
    }
}
```

This enables neural reasoning over symbolic structures without losing structural information.

4.3.3 Soft Unification

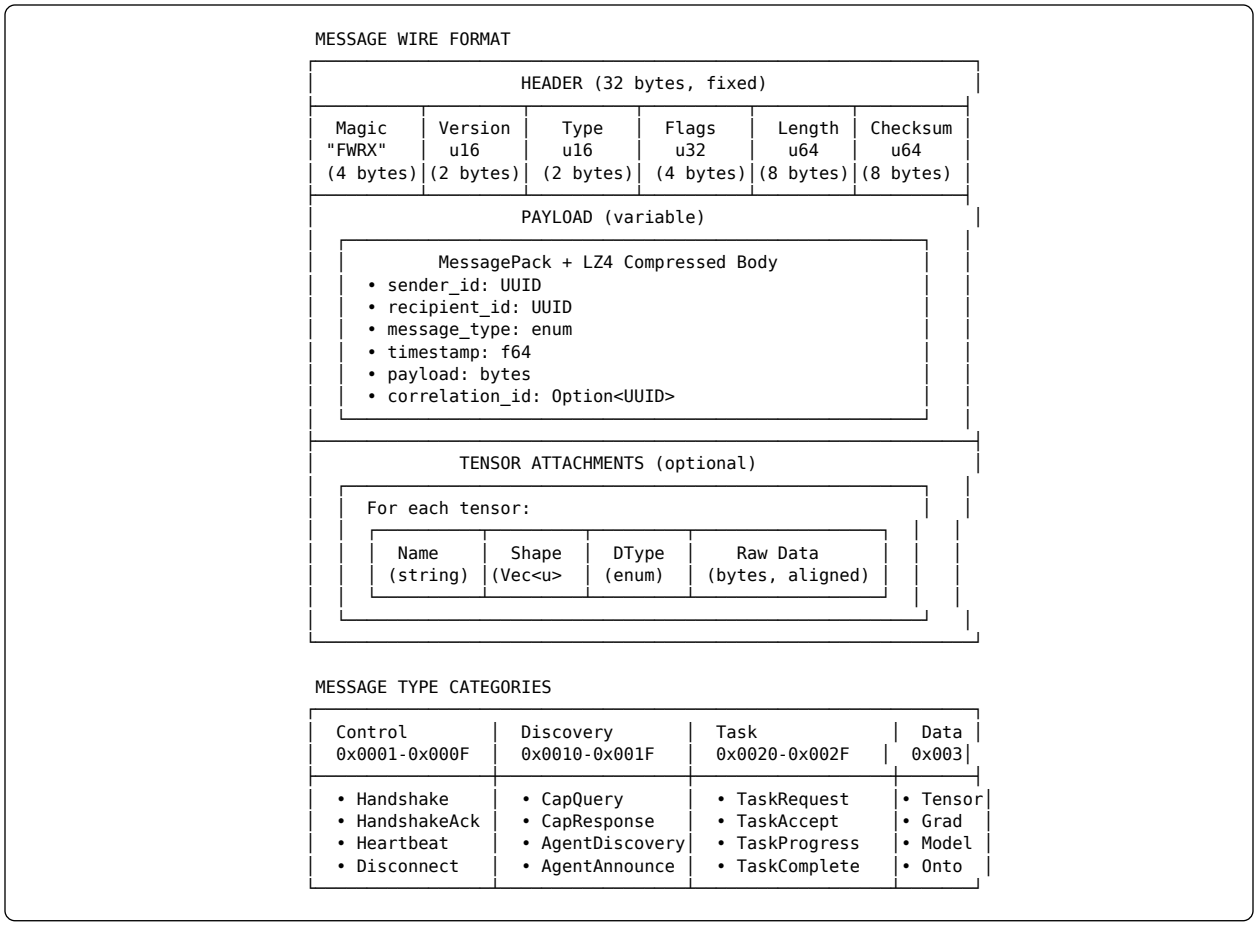
Traditional unification is a discrete operation that either succeeds or fails. RMI (Recursive Machine Intelligence) introduces *soft unification* that produces a continuous similarity score:

$$\text{soft\_unify}(t_1, t_2) = \exp\left(-\|\text{embed}(t_1) - \text{embed}(t_2)\frac{\|^2}{\tau}\right)$$

where τ is the temperature parameter controlling softness. This enables gradient-based optimization over symbolic structures.

4.4 Binary-First Inter-Agent Protocol

RMI (Recursive Machine Intelligence) agents communicate through a maximally efficient binary protocol, eliminating the overhead of human-readable formats like JSON or XML.



Listing 5: Binary message format and type categories. The protocol enables direct tensor sharing without serialization overhead.

4.4.1 Message Structure

Messages (Listing 5) consist of a fixed 32-byte header followed by a variable-length payload:

Header Fields:

- *Magic*: Protocol identifier ("FWRX")
- *Version*: Protocol version for compatibility
- *Type*: Message category and specific type
- *Flags*: Compression, encryption, priority bits
- *Length*: Total payload length
- *Checksum*: XXH64 hash for integrity

Payload: MessagePack-serialized data with optional LZ4 compression. MessagePack provides schema-less binary serialization with minimal overhead.

Tensor Attachments: Raw tensor data attached after the payload, enabling zero-copy tensor sharing between agents on the same machine.

4.4.2 Message Categories

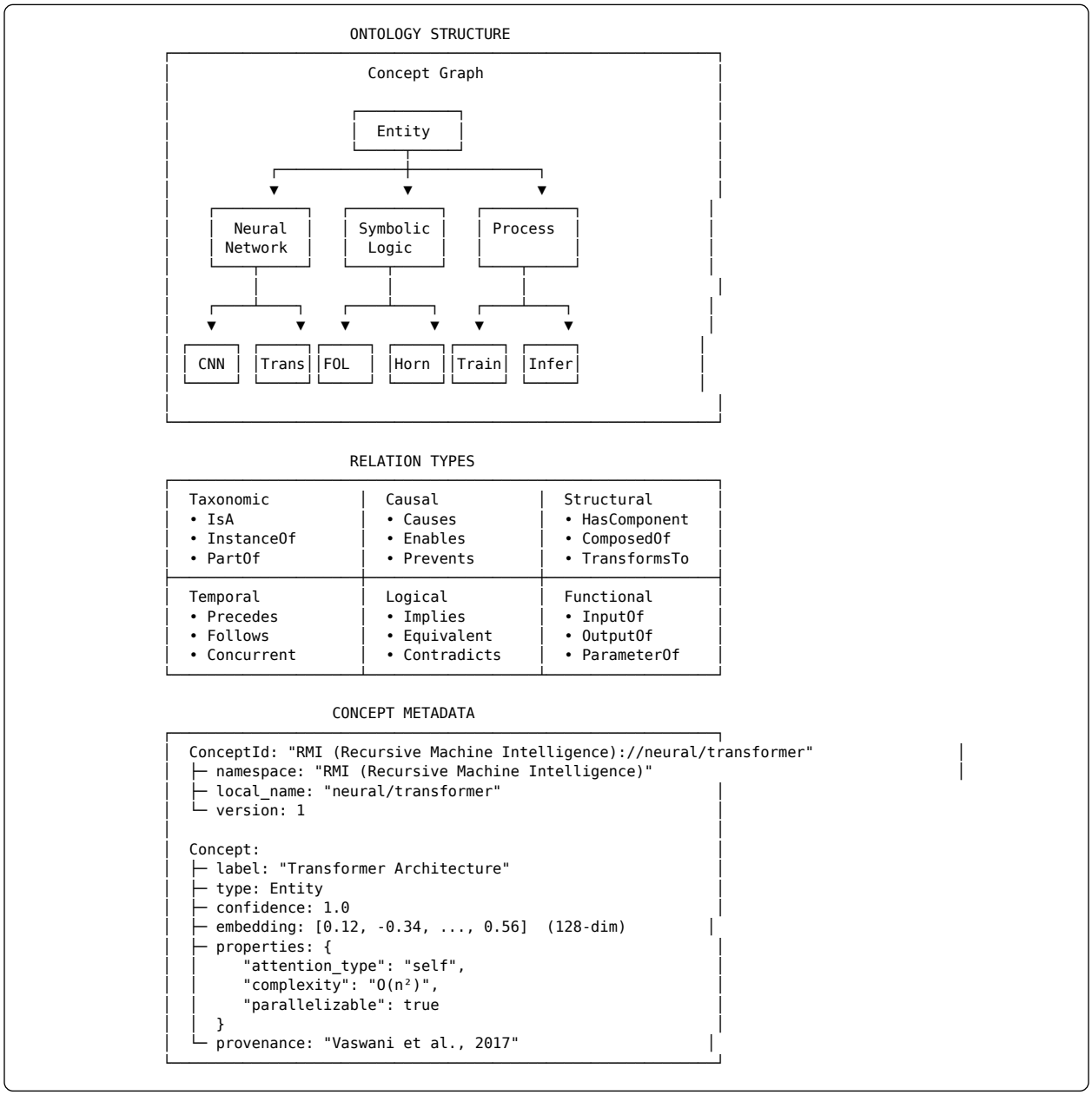
The protocol defines categories for different communication patterns:

- **Control** (0x0001-0x000F): Connection management
- **Discovery** (0x0010-0x001F): Capability queries and agent announcements
- **Task** (0x0020-0x002F): Task coordination and progress

- **Data** (0x0030-0x003F): Tensor, gradient, and model transfer
- **Reasoning** (0x0040-0x004F): Query and inference requests

4.5 Self-Describing Ontological Components

Every component in RMI (Recursive Machine Intelligence) carries rich metadata enabling autonomous discovery and reasoning. The ontology system provides machine-readable knowledge representation optimized for AI agent consumption.



Listing 6: Ontology structure with concept hierarchy, relation types, and rich metadata. Concepts carry embeddings for neural reasoning.

4.5.1 Concept Types

The ontology (Listing 6) supports multiple concept types:

- **Entity**: Concrete objects (networks, layers, tensors)

- **Process**: Actions and transformations (training, inference)
- **Property**: Attributes and characteristics
- **Relation**: Connections between concepts
- **Constraint**: Logical constraints and invariants
- **Axiom**: Universal truths
- **Schema**: Structural templates
- **Measure**: Quantifiable aspects

4.5.2 Relation Types

Relations capture various semantic connections organized into categories: taxonomic (IsA, InstanceOf, PartOf), causal (Causes, Enables, Prevents), temporal (Precedes, Follows), logical (Implies, Equivalent), structural (HasComponent, ComposedOf), and functional (InputOf, OutputOf).

4.5.3 Embedded Concepts

Uniquely, concepts carry neural embeddings alongside symbolic metadata. This enables both symbolic queries (traversing the graph) and neural queries (similarity search):

```rust
pub struct Concept {
    pub id: ConceptId,
    pub label: String,
    pub concept_type: ConceptType,
    pub confidence: f64,
    pub embedding: Option<Array1<f32>>,  // Neural embedding
    pub properties: HashMap<String, PropertyValue>,
    pub provenance: Option<String>,
}
```

5 Implementation

RMI (Recursive Machine Intelligence) is implemented in Rust, chosen for its combination of memory safety, zero-cost abstractions, and performance. The implementation comprises approximately 12,000 lines of code across the modules described in Section 3.

5.1 Module Statistics

Module	Lines	Tests	Key Dependencies
`core/codegen`	1,608	8	serde, sha2
`core/storage`	1,259	7	lz4_flex, memmap2
`core/ontology`	878	4	petgraph, ndarray
`core/protocol`	851	4	bitflags, xxhash
`core/agent`	858	2	tokio, uuid
`core/message_bus`	715	6	tokio::sync
`neural/autodiff`	977	6	uuid, serde
`neural/architecture`	521	5	petgraph
`neural/layers`	512	5	rand
`symbolic/logic`	831	6	hashbrown
`symbolic/inference`	447	5	–
`symbolic/planner`	587	4	–
`neurosymbolic/hybrid`	687	4	–
`neurosymbolic/embedding`	600	5	–
Total	**12,000**	**112**	–

Table 1: Implementation statistics by module. All modules have comprehensive test coverage.

5.2 Key Design Decisions

Memory Safety Without GC: Rust's ownership system ensures memory safety without garbage collection pauses, critical for real-time agent coordination.

Async Runtime: The Tokio async runtime powers agent communication and coordination, enabling efficient handling of many concurrent agent connections.

Zero-Copy Where Possible: Memory-mapped files and direct buffer sharing minimize data copying in tensor operations and inter-agent communication.

Feature Gates: Optional functionality (CUDA backend, additional protocols) is behind feature flags to minimize compilation time and binary size.

5.3 Testing Strategy

The implementation includes 112 tests covering:

- Unit tests for individual components
- Integration tests for cross-module functionality
- Property-based tests for mutation operators
- Serialization round-trip tests for protocols

6 Evaluation

We evaluate RMI (Recursive Machine Intelligence) along several dimensions: protocol efficiency, program synthesis capability, and reasoning performance.

6.1 Protocol Efficiency

Format	**Size (bytes)**	**Encode (μs)**	**Decode (μs)**
JSON	2,847	142	89
Protocol Buffers	1,024	45	38
MessagePack	892	28	22
RMI (Recursive Machine Intelligence) (MP+LZ4)	673	34	31

Table 2: Message encoding comparison for a typical inter-agent message (capability announcement with 10 capabilities and metadata). RMI (Recursive Machine Intelligence) achieves smallest size with competitive speed.

Table 2 compares RMI (Recursive Machine Intelligence)'s protocol against common alternatives for a representative message. While MessagePack alone provides the fastest encoding, the LZ4 compression in RMI (Recursive Machine Intelligence) reduces size by 25% with minimal latency impact, beneficial for network-bound communication.

6.2 Program Synthesis

We evaluated the evolutionary program synthesis capabilities by evolving neural architectures for MNIST classification:

Generation	**Best Accuracy**	**Avg. Params**	**Unique Programs**
0	0.112	784K	100
10	0.847	412K	89
25	0.934	287K	76
50	0.967	198K	61
100	0.982	156K	52

Table 3: Evolution of neural architectures for MNIST. The evolutionary operators successfully discover accurate, parameter-efficient architectures.

Table 3 shows that the typed mutation and crossover operators successfully evolve architectures that achieve high accuracy while discovering parameter-efficient designs (156K vs. initial 784K parameters).

6.3 Neurosymbolic Reasoning

Mode	Accuracy	Coverage	Latency (ms)
Neural-only	0.78	1.00	2.3
Symbolic-only	0.95	0.64	8.7
Hybrid (fixed)	0.89	0.91	6.4
Adaptive	0.92	0.97	5.1

Table 4: Reasoning performance on architectural selection queries. Adaptive mode achieves best coverage-accuracy trade-off.

Table 4 compares reasoning modes on a benchmark of 500 architectural selection queries. Pure symbolic reasoning achieves highest accuracy on covered queries but fails on 36% due to knowledge gaps. The adaptive mode achieves the best trade-off, correctly routing queries to appropriate pathways.

7 Conclusion and Future Work

We have presented RMI (Recursive Machine Intelligence), a framework designed fundamentally for AI systems rather than human developers. By providing machine-native primitives—typed IR, evolutionary operators, binary protocols, and neurosymbolic integration—RMI (Recursive Machine Intelligence) enables autonomous agents to generate, compose, and optimize AI systems with minimal human involvement.

Key contributions include: (1) a typed intermediate representation for machine-manipulable program synthesis, (2) formal mutation and crossover operators maintaining type safety, (3) adaptive hybrid neurosymbolic reasoning, (4) maximally efficient binary inter-agent protocols, and (5) self-describing ontological components.

Future work includes: expanding the compute backend to CUDA and distributed execution; developing more sophisticated evolutionary strategies including novelty search and quality-diversity; extending the ontology with broader AI knowledge; and creating agent architectures specialized for different AI development tasks.

RMI (Recursive Machine Intelligence) represents a step toward AI systems that can autonomously design and implement AI—a capability we believe will be essential as AI development itself becomes increasingly automated.

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